

Atom Basics:

Learning Intentions

- We will work independently to recall some of the main concepts of atomic theory
- We will discuss our current understanding of atomic theory and fill in any current gaps
- We will learn some of the important language used in science to talk about different types of particles

Curricular Competencies

- **Q&P:** I will demonstrate a sustained intellectual curiosity by testing my current knowledge of atomic theory, and filling in any gaps by asking good questions during discussion

Isotopes and AAM

Learning Intentions

- We will discuss "ISOTOPES"
- We will discuss "relative mass"
- We will draw and discuss the different uses for isotopes.
- We will cover the details of "isotopes" and understand how our knowledge of isotopes affects the "average atomic mass" of each element.
- We will see how scientific inventions help us understand the world on a deeper level

Curricular Competencies

- **PAI:** Use knowledge of scientific concepts to draw conclusions that are consistent with evidence

Bohr Diagrams

Learning Intentions

- We will review our previous understandings and conception's of what "atoms" are
- We will review Bohr Diagrams, and some of the limitations which come with using them.

Curricular Competencies

- **E:** We will evaluate the validity and limitations of our current understanding of atoms.

COULOMBIC POTENTIAL ENERGY & IONIZATION ENERGY.

Chem Activity 3 Learning Intentions

- We will use data to understand the relationships between charge and distance of two particles
- We will use data to understand the relationship between "ionization energy" and "coulombic potential"
- We will solve problems to further solidify our understanding of the key equation

Chem Activity 3 Success Criteria

PAI: I can analyze how changes in nuclear and electron charge (q_1 and q_2) and the distance between them (d) affect the ionization energy by applying Coulomb's Law, and use this relationship to estimate relative ionization energies across atoms.

INTRODUCING THE SHELL MODEL

Chem Activity 4 Learning Intentions

- We will discover what the "first ionization" energy is
- We will use data and hypothesis to compare atoms with different characteristics and IEs

Chem Activity 4 Success Criteria

- **QP:** I can make informed predictions about the relative first ionization energies of two hypothetical atoms by reasoning through how atomic structure — such as nuclear charge and electron distance — influences the energy required to remove an electron.

SHELL MODEL CONTINUED

Chem Activity 5 Learning Intentions

- We will learn about valence vs. inner shell electrons
- We will discover the concept of "core charge" using data and diagrams
- We will connect the concepts of core charge and ionization energy
- We will use data to strengthen our understanding of core charge and ionization energy

Chem Activity 5 Success Criteria

- **QP:** I can predict how the number of valence electrons influences the ionization energy of an atom, using ideas about electrostatic forces and atomic structure.
- **PC:** I can calculate the core charge of any atom in the first three periods of the periodic table using periodic trends and atomic structure.

- **PAI:** I can analyze how valence electrons and core charge relate to each other by identifying patterns across periods and groups in the periodic table.
- **E:** I can evaluate how core charge affects ionization energy by connecting changes in nuclear attraction to energy required for electron removal, considering atomic size and effective nuclear charge

ATOMIC RADIUS

Chem Activity 6 Learning Intentions

- We will learn about the concept of atomic radius
- We will connect core charge to atomic radius
- We will learn how adding or removing electrons affects the atomic radius

Chem Activity 6 Success Criteria

QP: I can compare two atoms and hypothesize which one has a larger atomic radius based on their position in the periodic table and their core charge.

PAI: I can analyze how gaining or losing electrons affects the radius of an ion by comparing their core charge and electron configuration.

Light Waves Electrons:

Learning Intentions

- We will review the properties of waves and light
- We will discover how light and energy are related
- We will calculate the amount of energy associated with different light sources
- We will see how we can measure IE using photons

Curricular Competencies:

- Q&P: Demonstrate curiosity about how light's properties relate to energy.
- PAI: Describe relationships between variables, perform calculations.
- A&I: Use multiple strategies to solve applied and conceptual problems.
- C: Construct evidence-based explanations using scientific language.

Photoelectron Spectroscopy

ChemActivity 8 Learning Intentions

- We will develop an understanding of the relationship between Ionization energy, energy of a photo, and kinetic energy, using the equation $IE = E_{\text{photon}} - KE$
- We will analyze graphs to understand how the electrons in an atom are organized
- We will begin to see how electrons are organized in an atom based on "quantized energy levels"

ChemActivity 82

- **PAI:** I can analyze and interpret spectral data or ionization energy graphs to identify unknown elements by comparing patterns and values.

- **PAI:** I can construct a model showing the relationship between energy levels and electron configuration in Bohr diagrams.
- **C:** I can model and explain, using diagrams or mental images, how photons interact with electrons to cause ionization.

Chem Activity 10, 11,12 Learning Intentions

- We will learn about shells, subshells, and orbitals
- We will discover how electrons are arranged in specific, quantized places within an atom
- We will link concepts from previous chem activities to synthesize our understandings
- We will use PES to hypothesize the electron configuration of atoms
- We will discuss Hund's rule, Pauli principal, and the Aufbau principal and how "spin" relates to electron configurations

Chem Activity 10, 11,12 Success Criteria

C: I can use scientific terminology to explain the structure of an atom in terms of shells, subshells, and orbitals as part of electron configuration.

PAI: I can construct and interpret full, short-hand, and orbital diagrams for electron configurations of atoms.

E: I can critically evaluate the strengths and limitations of previous models (like the Bohr diagram and planetary model) and how they can or cannot apply PES, Aufbau, Hund's rule, and Pauli exclusion when applied to real atomic systems.

Periodic Table Trends

Learning Intentions

- We will discuss the patterns and trends which exist on the periodic table
- We will learn about terms such as affinity, ionization, and electronegativity
- We will relate the trends to previous learnings such as radius and ionization energy
- We will see how the trends change as electrons are gained or lost by atoms

Success Criteria

QP: I can make predictions about chemical properties or reactivity based on multiple trends, including atomic radius and ionization energy, and support those predictions using prior learning.

PAI: I can identify and explain periodic trends (e.g., atomic radius, electronegativity, ionization energy) by analyzing their patterns across periods and down groups on the periodic table and justify those patterns based on their atomic configuration and associated properties.

C: I can accurately define and use key terms such as electron affinity, ionization energy, and electronegativity when explaining periodic trends or comparing elements.

UNIT 1 Atomic Structure and Properties

TOPIC 1.5 Atomic Structure and Electron Configuration

SAP-1 Atoms and molecules can be identified by their electron distribution and energy.

SAP-1.A Represent the electron configuration of an element or ions of an element using the Aufbau principle. ▼ Show details

SAP-1.A.1 The atom is composed of negatively charged electrons and a positively charged nucleus that is made of protons and neutrons. ▼ Show details

SAP-1.A.2 Coulomb's law is used to calculate the force between two charged particles. EQN : $F_{\text{coulombic}} \propto \frac{q_1 q_2}{r^2}$ ▼ Show details

SAP-1.A.3 In atoms and ions, the electrons can be thought of as being in "shells (energy levels)" and "subshells (sublevels)," as described by the electron configuration. Inner electrons are called core electrons, and outer electrons are called valence electrons. The electron configuration is explained by quantum mechanics, as delineated in the Aufbau principle and exemplified in the periodic table of the elements. ▼ Show details

SAP-1.A.4 The relative energy required to remove an electron from different subshells of an atom or ion or from the same subshell in different atoms or ions (ionization energy) can be estimated through a qualitative application of Coulomb's law. This energy is related to the distance from the nucleus and the effective (shield) charge of the nucleus. ▼ Show details

TOPIC 1.6 Photoelectron Spectroscopy

SAP-1 Atoms and molecules can be identified by their electron distribution and energy.

SAP-1.B Explain the relationship between the photoelectron spectrum of an atom or ion and: The electron configuration of the species. The interactions between the electrons and the nucleus. ▼ Show details

SAP-1.B.1 The energies of the electrons in a given shell can be measured experimentally with photoelectron spectroscopy (PES). The position of each peak in the PES spectrum is related to the energy required to remove an electron from the corresponding subshell, and the height of each peak is (ideally) proportional to the number of electrons in that subshell. ▼ Show details

TOPIC 1.7 Periodic Trends

SAP-2 The periodic table shows patterns in electronic structure and trends in atomic properties.

SAP-2.A Explain the relationship between trends in atomic properties of elements and electronic structure and periodicity. ▼ Show details

SAP-2.A.1 The organization of the periodic table is based on the recurring properties of the elements and explained by the pattern of electron configurations and the presence of completely or partially filled shells (and subshells) of electrons in atoms. ▼ Show details

SAP-2.A.2 Trends in atomic properties within the periodic table (periodicity) can be qualitatively understood through the position of the element in the periodic table, Coulomb's law, the shell model, and the concept of shielding/effective nuclear charge. These properties include: a. Ionization energy b. Atomic and ionic radii c. Electron affinity d. Electronegativity.

SAP-2.A.3 The periodicity (in SAP-2.A.2) is useful to predict /estimate values of properties in the absence of data. ▼ Show details

TOPIC 1.8 Valence Electrons and Ionic Compounds

SAP-2 The periodic table shows patterns in electronic structure and trends in atomic properties.

SAP-2.B Explain the relationship between trends in the reactivity of elements and periodicity. ▼ Show details

SAP-2.B.1 The likelihood that two elements will form a chemical bond is determined by the interactions between the valence electrons and nuclei of elements. ▼ Show details

SAP-2.B.2 Elements in the same column of the periodic table tend to form analogous compounds. ▼ Show details

SAP-2.B.3 Typical charges of atoms in ionic compounds are governed by their location on the periodic table and the number of valence electrons. ▼ Show details